

EXAM 7 – SPRING 2013

16. (1 point)

An insurer with a large, stable book of personal auto insurance is building an internal model for solvency. The actuary building the model assumes that the loss ratios are positively correlated from one year to the next.

a. (0.5 point)

Provide two reasons why this assumption is reasonable.

b. (0.5 point)

Describe how the model should incorporate this assumption.

CONTINUED ON NEXT PAGE

Question 16 Sample Answer

Solution 1

- a) Any two of the following common responses. Others were accepted if they were explained well and made sense.
- i. It is a stable book, so you will have the same policyholders year after year, so the book should perform similarly over time.
 - ii. It is a large book, so it has high credibility and you would not expect it to have large swings in loss ratio from year to year.
 - iii. Personal auto is highly regulated, so it is difficult to get significant rate changes and so rate levels will not fluctuate wildly.
 - iv. Low catastrophe exposure, which would lead to more consistent results and positively correlated loss ratios.
 - v. The underwriting cycle causes rate levels to be low for a period of time, and then high for a period of time. If the years are all in the same period of the cycle, the rate level would be consistent and this would make it more likely that the loss ratios would be positively correlated across years.
- b) To address this assumption:
- i. Input for expected rating environment
 - ii. Volatility around the mean selected
 - iii. Volatility should increase over new projection period
 - iv. Appropriate correlation between premium rating movements for different projection years and between different classes

Examiner Comment

We accepted other answers for partial or full credit if they were well explained and made sense. We received a wide variety of answers on this question, as it was very open ended. A common way to lose points was to not give enough explanation, such as simply stating that you would build correlation into the model, but not explaining how. This was seen as restating the question, and no credit was given unless there was more detail.